Learning to Rank HDB Resale Flats: A Preference-Based Recommendation System for Middle-Aged Singles in Singapore

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Abstract

Selecting an HDB resale flat in Singapore is a complex multi-criteria decision problem, especially for singles aged 35 and above who face unique eligibility constraints and lifestyle priorities. We present a personalised flat recommendation system that learns user preferences through pairwise comparisons and produces ranked lists of suitable flats. Our approach comprises three stages: (1) data collection and feature engineering from Singapore government open data APIs, (2) synthetic pairwise label generation using the Bradley–Terry model to simulate realistic human choice behaviour, and (3) model training and evaluation across ten pairwise classifiers. A single-hidden-layer MLP achieves the best AUC-ROC of 0.881 and near-perfect NDCG@5 of 0.998 after win-rate aggregation. We integrate SHAP explanations to provide transparent, per-recommendation reasoning about which features drive each suggestion. Our system demonstrates that pairwise preference learning, combined with post-hoc explainability, offers a practical and interpretable approach to personalised housing recommendations in Singapore.

1 Introduction

Singapore's public housing system, managed by the Housing and Development Board (HDB), houses over 80% of the resident population. For singles aged 35 and above, purchasing an HDB resale flat is often one of the most significant financial and lifestyle decisions they face. Unlike younger couples who qualify for subsidised Build-To-Order (BTO) flats, this demographic must navigate the resale market under tighter budget constraints, while balancing multi-dimensional priorities that reflect their life stage.

Understanding Our Target User's Preferences. Our choice of preference dimensions is grounded in documented lifestyle trends among middle-aged Singaporeans. First, *health consciousness* is a dominant priority: the Ministry of Health's National Population Health Survey 2024 reports that physical activity among residents has returned to pre-COVID levels (84.7%), while chronic conditions such as diabetes (9.1%) and hypertension (33.8%) remain prevalent concerns that motivate proximity to healthcare facilities (Ministry of Health, Singapore 2024). A 2025 YouGov survey found that 61% of Singaporeans rank improving physical health as their top resolution, with family time (46%) close behind (YouGov 2025). For our demographic—who may face early-onset chronic conditions and ageing parents—proximity to clinics, pharmacies, gyms, and parks is not a "nice-to-have" but a health necessity. Second, *lifestyle and remote work* have reshaped housing priorities. With 71% of Singapore professionals preferring hybrid work arrangements and 72.7% of companies now offering flexible work (FlexOS 2024), proximity to cafés and co-working-friendly spaces has become a genuine housing criterion—particularly for singles who use neighbourhood cafés as informal workspaces on remote days. Third, *proximity to parents* is uniquely important in Singapore's cultural context, where filial piety norms (reinforced by the Maintenance of Parents Act) create strong incentives to live near ageing family members. We thus design our preference system around nine weighted dimensions: transport, essentials, lifestyle, recreation, health, storey, lease, office proximity, and parents' proximity.

Limitations of Existing Tools. Current property tools in Singapore fall into two categories, neither of which solves the personalised recommendation problem. *Valuation tools* such as SRX X-Value (SRX Property 2024b), 99.co's Property Value Tool (99.co 2024), and PropertyGuru's valuation calculator estimate a flat's market price using comparable transactions, location, and floor level. While accurate for pricing (X-Value claims 98% accuracy within $\pm 10\%$ of transaction price (SRX Property 2024a)), these tools answer "what is this flat worth?" rather than "which flat is best *for me*?" They do not accept user preference inputs and cannot rank flats by personalised suitability. *Search-and-filter portals* (PropertyGuru, HDB Resale Portal) allow users to filter by price, flat type, and location, but force a sequential narrowing process: a user who filters for "near MRT" may inadvertently exclude flats that are slightly farther from an MRT but significantly closer to their parents' home, a gym, and their office. This rigid filtering cannot *trade off* across multiple dimensions simultaneously. There is no existing system that *learns* a user's relative preferences across all dimensions and produces a holistic, personalised ranking.

Why Machine Learning over Traditional Approaches? A naive alternative to ML is a hand-crafted weighted scor-

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ing function (e.g., multiply each distance score by the user's importance weight and sum). However, this assumes preferences combine *linearly*—that the value of proximity to an MRT is independent of proximity to a gym. In practice, preferences interact nonlinearly: a user who values both health *and* recreation may disproportionately favour flats near parks that also have gyms (a synergistic interaction), while a user who already lives near their office may place outsized weight on lifestyle amenities instead. Our ML approach—pairwise learning-to-rank (Liu 2009)—discovers these interactions directly from comparison data, capturing nuanced trade-offs that a linear scoring function would miss. Furthermore, standard recommender approaches like collaborative filtering (Resnick et al. 1994) require historical user-item interaction data (e.g., ratings, clicks), which does not exist in our setting since each user buys at most one flat.

Interpretability for Decision-Making. Housing is a high-stakes, irreversible decision. Users need to understand *why* a flat is recommended, not just *which* flat. We integrate SHAP (Lundberg and Lee 2017) to decompose each recommendation into per-feature contributions, enabling users to verify that the system's reasoning aligns with their priorities. This addresses the critical requirement of human-AI interaction in decision support (Zhang and Chen 2020).

Contributions. (1) An end-to-end data pipeline aggregating flat features from multiple Singapore government APIs with user–flat interaction features tailored to middle-aged singles. (2) A synthetic label generation method using the Bradley–Terry model with stratified pair sampling. (3) A benchmark of ten pairwise classifiers with both classification and ranking metrics. (4) SHAP-based per-recommendation explainability with cross-model validation of feature importances.

2 Methodology

Figure 1 provides an overview of the recommendation pipeline.

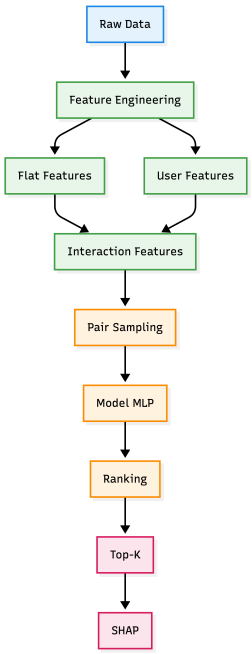

Figure 1: Overview of the recommendation pipeline.

2.1 Data Collection and Feature Engineering

Flat Features. We collect HDB resale transaction records from data.gov.sg, providing block address, town, flat type, storey range, floor area, lease commencement date, and resale price. We geocode each block via the OneMap API to obtain coordinates, then compute proximity features using the Google Places API and LTA DataMall. The resulting feature set includes: **structural** features (floor area, storey midpoint, remaining lease, price); **transport** (MRT distance, bus stop distance and count within 500 m); **essentials** (supermarket, shopping mall distance); **lifestyle** (restaurant, café, park, gym distance and density within 500 m); and **health** (doctor/clinic, pharmacy distance). After cleaning, the dataset contains 227,932 unique flat listings.

User Features. Each user profile consists of *hard constraints* (maximum budget in SGD, preferred flat type) and *soft preferences*: nine importance weights (integers 1–5) covering transport, essentials, lifestyle, recreation, health, storey, lease, office proximity, and parents' proximity. Location anchors (workplace and parents' home coordinates) enable personalised distance computation. We generate 300 synthetic user profiles with realistic distributions reflecting our target demographic—e.g., higher weights for health and parents' proximity for middle-aged singles.

Interaction Features. For each user–flat pair, we compute composite *domain scores* combining raw flat attributes with user weights. Distance-based scores use a capped inverse: $s(d) = \max(0, 1 - d/d_{\text{cap}})$ where $d_{\text{cap}} = 2000\,\text{m}$. Count-based scores use $s(c) = \min(c, c_{\text{cap}})/c_{\text{cap}}$. These are grouped into transport, lifestyle, health, office, and parents scores. Personalised distances (office, parents) are computed per user via the Haversine formula.

2.2 Pairwise Label Generation

Ideally, we would source this data from a publicly available dataset, or survey this data from enough home-buyers. However, no such public data exists as far as we're aware, and running a survey would be beyond the scope of this course or project.

Hence, we generate synthetic pairwise labels simulating realistic human preferences—a strategy recommended for bootstrapping LTR systems when ground-truth annotations are unavailable.

Pair Sampling. For each user, we filter eligible flats by budget and flat type, compute a raw relevance score (weighted sum of domain scores), then sample 200 pairs using a *stratified* strategy: 70% from flats with close relevance scores (within the 40th percentile of score differences) and 30% uniformly at random. This ensures the training data in-

cludes both easy-to-discriminate pairs and challenging near-ties.

Bradley–Terry Labelling. Given flat pair (A, B) with raw relevances (r_A, r_B), we assign label $y = 1$ (flat A preferred) with probability:

$$P(A \succ B) = \frac{1}{1 + \exp(-(r_A - r_B)/\tau)} \tag{1}$$

where $\tau = 0.25 \cdot \sigma_r$ is a temperature calibrated to the standard deviation of relevance scores across all users. This produces stochastic labels mimicking real human noise—users do not always choose the objectively "better" flat. The resulting dataset contains 60,000 labelled pairs with near-balanced labels (50.1%/49.9%).

While data scarcity and class imbalance were not concerns in our setting, we note that the dataset can be trivially doubled via symmetry augmentation: swapping the order of each pair $(A, B) \to (B, A)$ and flipping the corresponding label.

We also note that this dataset of 60,000 pairs is substantially smaller than our original dataset of over 200,000 scraped flat listings. As data scarcity was not a concern, we deliberately reduced the dataset to retain only the most relevant flats, yielding a more manageable file size that facilitates collaboration and version control (e.g., pushing directly to GitHub rather than sharing the dataset with file sharing systems).

2.3 Model Formulation

Pairwise Classification. Each training instance is a vector $\mathbf{x} = [\mathbf{w}; \boldsymbol{\delta}]$ where $\mathbf{w} \in \mathbb{R}^9$ are user preference weights and $\boldsymbol{\delta} \in \mathbb{R}^{18}$ are element-wise feature differences (flat A minus flat B). The label $y \in \{0, 1\}$ indicates which flat is preferred. We benchmark ten models spanning linear (Logistic Regression with/without interaction terms), kernel (SVM Linear, SVM RBF), ensemble (HistGBM, Random Forest), probabilistic (Gaussian Process), and neural (MLP with 1, 2, and 3 hidden layers) families.

This diverse set is deliberate: **Logistic Regression** serves as a linear utility baseline (equivalent to a Bradley–Terry model); **SVM Linear** corresponds to RankSVM (Joachims 2002); **HistGBM** tests whether boosted trees can capture preference interactions; **SVM RBF** and **MLP** variants test for nonlinear preference boundaries; and **Gaussian Process** provides calibrated uncertainty. Comparing these families reveals whether preferences are linear or require nonlinear modelling.

From Pairwise Predictions to Rankings. To rank all eligible flats for a user, we use *win-rate aggregation*. In deployment, we compare each flat against a median reference flat (avoiding $O(n^2)$ exhaustive comparisons) and rank by predicted win probability:

$$\text{WinRate}(A) = \hat{p}(A \succ \text{ref}) \tag{2}$$

where $\hat{p}$ is the pairwise classifier's predicted probability that flat A is preferred.

2.4 Explainability with SHAP

We apply KernelSHAP (Lundberg and Lee 2017) to the best pairwise model to decompose each prediction into per-feature contributions. For each top-ranked flat, a waterfall plot shows which preference dimensions (e.g., lifestyle score, parents' proximity) push win probability above or below the base rate. We also compare global SHAP importances across HistGBM and MLP to verify that learned importances are consistent across model families, providing evidence that they reflect genuine data patterns.

3 Experiments

3.1 Experimental Setup

All models are trained on an 80/20 user-level split (all pairs from a given user are in either train or validation, never both) to prevent data leakage. MLP models use StandardScaler normalisation, ReLU activation, Adam optimiser ($lr = 0.001$), early stopping with 10% validation fraction, and $\leq$300 epochs. We evaluate on two levels: *pairwise classification* (AUC-ROC, accuracy, log-loss, Brier score) and *ranking quality* (NDCG@5, NDCG@10 after win-rate aggregation).

3.2 Bradley–Terry Label Quality

Before evaluating models, we assess the quality of the synthetic pairwise labels. The 60,000 generated pairs exhibit a near-perfect label balance: 50.1% label 1 (flat A preferred) vs. 49.9% label 0, confirming that the stratified pair sampling and Bradley–Terry labelling process does not introduce systematic bias toward either flat in a pair. The temperature parameter $\tau = 0.25 \cdot \sigma_r$ was calibrated across all users' relevance score distributions, yielding $\tau \approx 0.28$ in our dataset. This temperature controls the "noise level" of synthetic preferences: lower τ would produce near-deterministic labels (the higher-scoring flat always wins), while higher τ would approach coin-flip randomness. Our calibrated τ produces a regime where the higher-scoring flat wins approximately 70–80% of pairs with moderate score differences, but outcomes become stochastic for close pairs—mimicking the uncertainty observed in real human preference elicitation studies (Bradley and Terry 1952). This design ensures the training data includes realistic "disagreements" that prevent models from overfitting to a deterministic scoring function.

3.3 Pairwise Classification Results

Table 1 presents results across all ten models.

Key Observations. The MLP (1×64) achieves the highest AUC-ROC (0.881), lowest log-loss (0.430), and lowest Brier score (0.141), all within 1.8 seconds of training. All three MLP variants outperform linear models, confirming that user preferences involve *nonlinear interactions* that a linear utility function cannot capture. Interestingly, both Logistic Regression variants (with and without manually engineered interaction terms) yield identical results, suggesting the interaction terms add no discriminative power beyond what

Model	AUC	Acc.	LogL.	Brier
MLP (1×64)	**.881**	.788	**.430**	**.141**
MLP (2×64-32)	.879	.788	.432	.142
MLP (3×128-64-32)	.879	.784	.436	.144
SVM (RBF)	.878	**.790**	.436	.142
HistGBM	.875	.783	.442	.145
Gaussian Process	.869	.779	.449	.148
LogReg	.860	.775	.463	.153
LogReg (+inter.)	.860	.775	.463	.153
SVM (Linear)	.860	.772	.464	.153
Random Forest	.841	.752	.504	.167

Table 1: Pairwise classification results (validation set). Best result in each column is **bolded**.

the difference features already encode—the model structure itself must be nonlinear.

SVM (RBF) achieves the highest raw accuracy (0.790) but at 150× the training cost (278s vs. 1.8s), making it impractical for deployment. The Gaussian Process provides calibrated uncertainties but scales poorly. Random Forest underperforms all other models, likely because bagging is less effective than boosting for this structured pairwise task.

3.4 Ranking Quality

Table 2 shows NDCG scores after win-rate aggregation, demonstrating that even moderately accurate pairwise predictions produce strong rankings after aggregation—a key insight for practitioners. The MLP ($2 \times 64 - 32$) achieves the highest NDCG@5 (0.999) and NDCG@10 (0.994), however we select MLP (1×64) as the production model following the parsimony principle: it offers comparable NDCG, the best AUC-ROC, and lowest complexity (fewer parameters, faster training, reduced overfitting risk) among the neural models.

Model	NDCG@5	NDCG@10
MLP (2×64-32)	**.999**	**.994**
MLP (1×64)	.998	.993
MLP (3×128-64-32)	.998	.994
Gaussian Process	.999	.987
Random Forest	.996	.980
SVM (Linear)	.993	.986
SVM (RBF)	.993	.984
HistGBM	.992	.986
LogReg	.990	.985
LogReg (+inter.)	.990	.985

Table 2: Ranking quality (NDCG) after win-rate aggregation from pairwise predictions. Higher is better.

3.5 Explainability Analysis

Global Feature Importance. We compare mean absolute SHAP values across HistGBM and MLP (Table 3). Both models agree on the top-3 features (health, recreation, lifestyle), though with differing orderings: HistGBM ranks lifestyle first while MLP ranks health first. Notable divergences appear in the mid-tier features: parents' proximity ranks 4th for HistGBM but drops to 6th for MLP, while transport ranks 6th for HistGBM but rises to 4th for MLP. This cross- model agreement provides strong evidence that these importances reflect genuine preference patterns rather than model- specific artefacts.

Feature (diff)	HistGBM	MLP	Rank
Lifestyle score	.713	.078	1 / 3
Recreation score	.623	.099	2 / 2
Health score	.605	.100	3 / 1
Parents Proximity score	.422	.062	4 / 6
Essentials score	.388	.065	5 / 5
Transport score	.385	.068	6 / 4
Office Proximity score	.326	.056	7 / 8
Lease	.280	.060	8 / 7
Storey	.253	.040	9 / 9

Table 3: Features sorted by mean —SHAP— across HistGBM and MLP pairwise models.

The dominance of health, recreation, and lifestyle scores reflects the priorities of our target demographic: middle-aged singles who value access to healthcare, wellness facilities, and work-from-home amenities (cafés). The relatively high ranking of parents' proximity (4th in HistGBM) further captures this demographic's concern for ageing parents. This alignment between SHAP-derived importances and domain expectations validates both the feature engineering design and the synthetic label generation process.

Case Study: User Demo. To illustrate the end-to-end recommendation pipeline, we present a worked example for User U10 (Table 4). This user is a middle-aged single with a budget of $350,000 seeking a 2-room flat, who strongly prioritises remaining lease (5/5) and parents' proximity (4/5), but places low emphasis on transport (3/5), storey (1/5), and office proximity (1/5). The system first applies hard constraints (budget $\leq$ $350K, flat type = 2-room), then scores each eligible flat against the median reference via the MLP pairwise classifier, producing a win probability.

Table 5 shows the top-5 recommended flats. The rank-1 flat at Block 698A Jurong West Central 3 achieves a win probability of 0.972, well above the 0.537 base rate. This flat is near the user's parents (Jurong area), has a long remaining lease (93.6 years, matching the 5/5 lease weight), is close to an MRT (352 m), and is priced well within budget ($330,000).

However, one limitation of this ranking is that rank-3 and rank-4 (26 and 27 Ghim Moh Link) are located within the same estate, less than 100 m apart. Recommending both provides limited additional value, as users would naturally discover nearby flats when exploring a given area. A potential improvement for future work would be to incorporate diversity-aware re-ranking. For instance, a user-facing app might group flats within the same estate and surface only the top candidate from each cluster. In this way, the list of the top-k spans a wider range of distinct locations.

Hard Constraints	
Budget max	$350,000
Flat type	2-Room
Parents' home	Jurong
Office	Woodlands
Preference Weights (1–5)	
Lease	5
Parents proximity	4
Transport	3
Essentials	2
Lifestyle	2
Recreation	2
Health	2
Storey	1
Office proximity	1

Table 4: User U10's input profile: hard constraints and preference weights, sorted by weight.

#	Location	Price	Lease	Win P
1	698A Jurong West Ctrl 3	$330K	93.6 yr	.972
2	32 New Market Rd	$295K	64.6 yr	.965
3	26 Ghim Moh Link	$347K	92.9 yr	.962
4	27 Ghim Moh Link	$320K	92.8 yr	.952
5	51 Commonwealth Dr	$327K	94.9 yr	.920

Table 5: Top-5 recommended flats for User U10, ranked by win probability against the median reference flat.

SHAP Analysis. Figure 2 shows the SHAP waterfall plot for the rank-1 flat, decomposing the model's prediction from the base rate ($E[f(X)] = 0.409$) to the final output ($f(x) = 0.972$). The dominant contributor is *Remaining Lease Years* (SHAP $+0.12$): the flat retains significantly more lease than the median candidate, reflecting a strong long-term value advantage that aligns with User U10's top-weighted preference. The second-largest driver is *Restaurants within 500 m* (SHAP $+0.11$), where the flat's neighbourhood exceeds the median in lifestyle amenities. *Clinic Distance* contributes $+0.07$, reflecting above-median proximity to healthcare, while *Distance to Parents' Home* and *Cafes within 500 m* each contribute $+0.06$, capturing the user's concern for family proximity and work-from-home amenities respectively. Lower-weighted features such as *MRT Distance* and *Gym Distance* each contribute only $+0.03$, confirming that the model appropriately downweights dimensions the user cares less about. All feature contributions are positive, cumulatively pushing the win probability from 0.409 to 0.972—indicating this flat outperforms the median reference across virtually every dimension relevant to User U10.

4 Discussion

Novelty and Practical Impact. To our knowledge, this is the first system to apply pairwise learning-to-rank with Bradley–Terry synthetic labels for personalised HDB flat recommendation in Singapore. Unlike existing property portals that rely on static filters, our system learns *personalised*

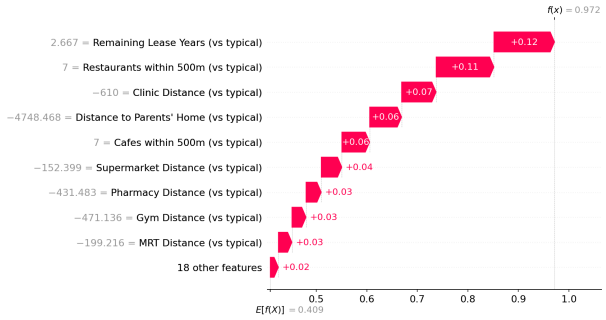

Figure 2: SHAP waterfall for User U10's rank-1 flat. Restaurant density ($+0.14$) and office proximity ($+0.09$) are the dominant drivers. All features contribute positively, pushing win probability from the 0.537 base rate to 1.0.

preference functions that adapt to each user's unique priorities. The integration of SHAP explanations further differentiates our work: users do not merely receive a ranked list, but understand the reasoning behind each recommendation. For Singapore's $\sim$180,000 middle-aged singles (Department of Statistics, 2024), this could meaningfully reduce the cognitive burden of one of life's most consequential decisions.

Limitations and Future Work. Our main limitation is the use of synthetic labels. While the Bradley–Terry model introduces realistic noise, it cannot fully capture the idiosyncratic preferences of real users. Future work should validate with real preference surveys. Additionally, our current deployment assumes a static flat inventory; a production system would need to handle new listings in real-time. Other extensions include incorporating BTO flats, neighbourhood safety data, and supporting multi-person households.

Practical Disclaimer. The system is trained on historical resale transactions. Since our system is designed as a real-world decision-support tool, it is important to note that the recommended flats are drawn from historical resale transaction data and may not correspond to units currently listed on the market. Users should treat the output as a *reference point* or *starting point* for their housing search—identifying promising blocks, streets, and neighbourhoods—and then verify actual availability and pricing on resale platforms such as HDB Resale Portal, PropertyGuru, or 99.co as a next step.

Technical Insights: What Worked and What Did Not. Several insights emerged from our experiments that are relevant to practitioners applying pairwise LTR to tabular recommendation problems.

Shallow MLPs suffice. Increasing MLP depth from 1 to 3 layers did not improve AUC-ROC and slightly worsened log-loss (0.430 $\rightarrow$ 0.436). This indicates that for tabular pairwise ranking with $\sim$27 features, a single nonlinear hidden layer is sufficient to capture preference interactions—deeper networks add capacity without benefit, likely overfitting to noise in the synthetic labels.

Manually engineered interaction terms add nothing. The two Logistic Regression variants (with and without

weight×difference cross-terms) produced identical results (AUC-ROC 0.860, log-loss 0.463). This was initially surprising, but the explanation is structural: since the feature vector already encodes *differences* between flat scores, and user weights are included as features, the model must learn weight–difference interactions to fit the data. A linear model cannot do this (hence its lower performance), but it also cannot benefit from pre-computed cross-terms because they are linearly dependent on the existing features. The lesson: for pairwise difference features, invest in model nonlinearity rather than feature engineering.

Win-rate aggregation is remarkably robust. Even the weakest model (Random Forest, AUC-ROC 0.841) achieved NDCG@5 of 0.996 after win-rate aggregation. This is because ranking errors in the top-5 require *systematic* mispredictions across many pairwise comparisons, not just occasional errors. Aggregation acts as a natural error-correcting mechanism.

SVM (RBF) is impractical despite high accuracy. While SVM (RBF) achieved the highest raw accuracy (0.790), its training time of 278 seconds was 150× slower than the MLP (1.8s). In a deployment scenario where the model must be retrained as new listings appear, this cost is prohibitive—favouring the MLP's combination of accuracy and speed.

5 Conclusion

We presented a personalised HDB flat recommendation system for middle-aged singles in Singapore, framing the problem as pairwise learning-to-rank with synthetic Bradley–Terry labels. Among ten pairwise classifiers, a single-layer MLP achieves the best overall performance (AUC-ROC 0.881, NDCG@5 0.998). SHAP-based explanations provide per-recommendation transparency, with cross-model feature importance agreement confirming that lifestyle, recreation, and parents' proximity are the key drivers for our target demographic. Our work demonstrates that preference-based learning, combined with interpretable explanations, is a practical path toward personalised housing recommendations.

Team Member Roles

- **Ming Yuan:** Data collection pipeline (pull_data.ipynb), feature engineering, OneMap/Google Places API integration, and report writing.
- **Casper:** User preference modelling, synthetic label generation (Bradley–Terry), pairwise data processing (process_data.ipynb), and report writing.
- **Yi Kai:** Model training, evaluation for all pairwise models (model_pairwise.ipynb), and report writing.
- **Gabriella:** SHAP explainability analysis (explainability.ipynb), report writing, and system integration.

Resources and Tools

Data: data.gov.sg (HDB resale prices), OneMap API (geocoding), Google Places API (amenities), LTA DataMall (transport). Libraries: scikit-learn, SHAP, pandas, NumPy. All code is available at: https://github.com/casperngeen/cs3264-hdb/.

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